Sayem Khan

Permanent Resident of the United States

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SUMMARY

Forward-deployed AI engineer. Digital Technologies Lead at AlchLight LLC (Rochester, NY), working directly with the lab's engineers, scientists and leadership to turn their needs into working software: control and automation for a femtosecond-laser system, AI tools for research, and DoD SBIR proposals as Technical Lead. Dual M.S. in Computer Science and Mathematics and a B.Sc. in Electrical Engineering; 5+ years of teaching and mentoring across three institutions. **3,300+ GitHub contributions in the last year**.

PROFESSIONAL EXPERIENCE

- **Digital Technologies Lead**, AlchLight LLC, Rochester, NY *(June 2026 – Present)*
  - **Laser system integration:** built the software that connects the lab's motion controller, femtosecond laser and 3D laser scanners into one system, deployed on the lab's isolated machines and tuned on site with the operators (C#/.NET, Python, gRPC).
  - **Scan-to-machining workflow:** desktop applications that take a 3D scan of a part, locate it and generate the machine program to process it.
  - **AI research assistant:** an LLM agent that searches the lab's paper library and the web for a research question and returns ready-to-test recipes, improving from the scientists' feedback.
  - **DoD SBIR** (Small Business Innovation Research): **Technical Lead** on two Phase I proposals submitted in 2026: a digital twin for the Defense Logistics Agency and AI-driven adaptive radar waveform design.
  - **Agile delivery:** every change issue-tracked, code-reviewed and CI-tested; weekly progress and KPI reports to leadership; IT infrastructure aligned to **CMMC** (DoD cybersecurity certification).

EDUCATION

- **M.S. in Computer Science & Engineering**, The State University of New York at Buffalo (SUNY Buffalo), Buffalo, NY. *(September 2021 - May 2024)*
- **M.S. in Mathematics**, Jahangirnagar University (JU), Dhaka, Bangladesh. *(January 2017 - March 2018)*
- **B.Sc in Electrical and Electronic Engineering**, Khulna University of Engineering & Technology (KUET), Khulna, Bangladesh. *(February 2012 - October 2016)*

ML RESEARCH & PROJECTS

- **Numerai Tournament: End-to-End ML System, Live Every Round**
  - **What it does:** Numerai is a hedge fund that crowdsources its stock-market models through a data-science tournament. This system trains, evaluates and selects models on its data and **submits predictions every round with no manual steps**.
  - Gradient-boosted trees and neural networks, validated **walk-forward** on time-ordered data to guard against look-ahead leakage.
  - Distributed **GPU training** (Slurm, Ray) and scheduled cloud serving (Modal), with tested, CI-gated code.
- **Orchestra: Market-Data Platform for Numerai Signals** *(in development)*
  - Data service that collects market data for 90,000+ symbols across US and international markets into Qlib (Microsoft's open-source quant research platform) data warehouses built with DuckDB, served through a REST API (FastAPI).
  - Research service that runs LLM agents (Microsoft RD-Agent) to propose factors and models from papers and backtest them on that data, as candidate signals for Numerai Signals, Numerai's tournament for stock signals built from your own data.
  - Containerized services (Docker Compose).

- **Hangman AI Solver - Position-wise Neural Approach**
  - **Problem:** win Hangman on 1,000 games of words never seen in training (250K-word dictionary, black-box API, 6 wrong guesses allowed) against an 18% baseline.
  - **Meta-reinforcement learning (RL^2):** a recurrent policy that adapts within each game reached 45% on unseen words; the 26-letter action space limited sequential RL.
  - **Reframed the problem** after **BERT's masked language modeling**: predict the letter at every hidden position. A BiLSTM trained on 21M game states from 227K words, generated with 13 masking strategies, reached a 67.2% **official win rate,** 3.7× **the baseline**.
  - **Tested what else helps:** heuristic baselines (letter frequency, n-grams, ensembles) stayed at 15–25%; self-supervised contrastive training added 0.8 points at 4.7× the training time, so supervised training was kept.
  - Released as **Hangman AI Solver (v1.0)**, Zenodo, October 2025 (PyTorch Lightning, 5.2M parameters).

RESEARCH ARTICLES & ABSTRACTS

- Nur, N. B.; Bachmann, C. M.; Tyler, A. C.; Miller, A.; **Khan, S.**; Union, K. E.; Owens-Rios, W. A.; Bauch, T. D.; Lapszynski, C. S. **Mapping Soil Organic Matter, Total Carbon, and Total Nitrogen in Salt Marshes Using UAS-Based Hyperspectral Imaging.** *Journal of Geophysical Research: Biogeosciences*, 130, e2024JG008421, 2025.
- Nur, N.B.; Bachmann, C.M.; Tyler, A.C.; Miller, A.; Union, K.E.; **Khan, S.**; Owens-Rios, W.A.; Bauch, T.D. **Enhancing Wetland Soil Property Estimates with UAS Hyperspectral Imaging in Preparation for the NASA SBG Mission** *Proceedings of the AGU Fall Meeting, 2024*. Ed. American Geophysical Union. Washington, D.C.

TECHNICAL SKILLS

Languages:	Python, C++, Bash, SQL
GenAI / Agents:	LLM APIs (Anthropic, OpenAI), Anthropic SDK, MCP (Model Context Protocol), agent frameworks (Microsoft RD-Agent), RAG, Prompt Engineering
AI-Native Dev:	Claude Code, Cursor, GitHub Copilot
ML/DL:	PyTorch, PyTorch Lightning, scikit-learn, LightGBM, XGBoost, CatBoost, Transformers, Weights & Biases
Data Engineering:	market-data warehouses (Qlib, DuckDB), NumPy, pandas, Chroma
APIs & Integration:	REST APIs (FastAPI), gRPC, integration of vendor instruments and their SDKs
Cloud & Infra:	Docker, Modal (serverless cloud), Ray, Slurm (GPU cluster), GitHub Actions, Git, CircleCI, n8n, LaTeX
Security:	CMMC compliance, secure handling of API keys and passwords

EMPLOYMENTS

- **Volunteer Research Engineer**, RoCAL (Robotic Collaboration and Autonomy Lab), Rochester Institute of Technology (RIT), January 2026 – May 2026
- **Voluntary Research Assistant**, Chester F. Carlson Center for Imaging Science, Rochester Institute of Technology (RIT), September 2024 – December 2025
- **Graduate Teaching Assistant**, Computer Science, SUNY Buffalo, NY, September 2021 – May 2024
- **Graduate Teaching Fellow**, Mathematics, UMass Dartmouth, MA, September 2019 – May 2021
- **Adjunct Faculty (Mathematics)**, Department of Mechanical Engineering, Sonargaon Univ., Dhaka, June 2018 – May 2019
- **Assistant Engineer**, Power Transformer Division, Energypac Engineering, Dhaka, November 2016 – April 2018